

clusters and the distributed environment is managed via Apache Zookeeper within the implementation of Apache Storm. Research based algorithms and predictive analytics can be executed in parallel using Apache Storm.

MapReduce is a fault-tolerant distributed high performance computational framework which is used to process and evaluate huge amounts of data. MapReduce-like functions can be effectively implemented in Apache Storm using bolts, as the key logical operations are performed at the level of these bolts. In many cases, the performance of bolts in Apache Storm can outperform MapReduce.

Key advantages and features of Apache Storm

The key advantages and features of Apache Storm are that it is user friendly, free and open source. It is fit for both small and large scale implementations, and is highly fault tolerant and reliable. It is extremely fast, does real-time processing and is scalable. And it performs dynamic load balancing and optimisation using operational intelligence.

Installing Apache Storm and Zookeeper on a MS Windows environment

First, download and install Apache Zookeeper from <https://zookeeper.apache.org/>.

Next, configure and run Zookeeper with the following commands:

```
MSWindowsDrive:\> cd zookeeper-Version
MSWindowsDrive:\ zookeeper-Version> copy conf\zoo_sample.cfg
conf\zoo.cfg
MSWindowsDrive:\ zookeeper-Version> .\bin\zkServer.cmd
```

The following records are updated in *zoo.cfg*:

```
tickTime=2000
initLimit=10
syncLimit=5
dataDir= M$WindowsDrive:/zookeeper-3.4.8/data
```

Now, download and install Apache Storm from <http://storm.apache.org/> and set `STORM_HOME` to `MSWindowsDrive:\apache-storm-Version` following in environment variables.

Perform the modifications in *storm.yaml* as follows:

```
storm.zookeeper.servers:
- "127.0.0.1"
nimbus.host: "127.0.0.1"
storm.local.dir: "D:/storm/datadir/storm"
supervisor.slots.ports:
- 6700
- 6701
- 6702
- 6703
```

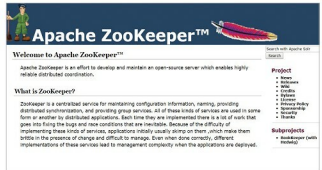


Figure 1: Official portal of Apache Zookeeper



Figure 2: Execution of commands to initialise Apache Zookeeper

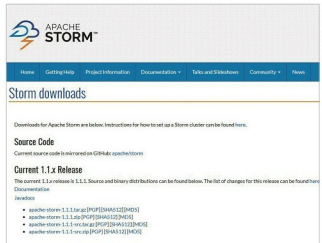


Figure 3: Download page of Apache Storm for multiple platforms



Figure 4: Execution of commands to initialise Apache Storm